DATA NETWORKS

Bhutan Insurance Limited

Early Warning System (EWS)

AI-Enabled Platform for Proactive Risk Detection & Mitigation

Data Networks Pvt. Ltd.

Version 1.1.0

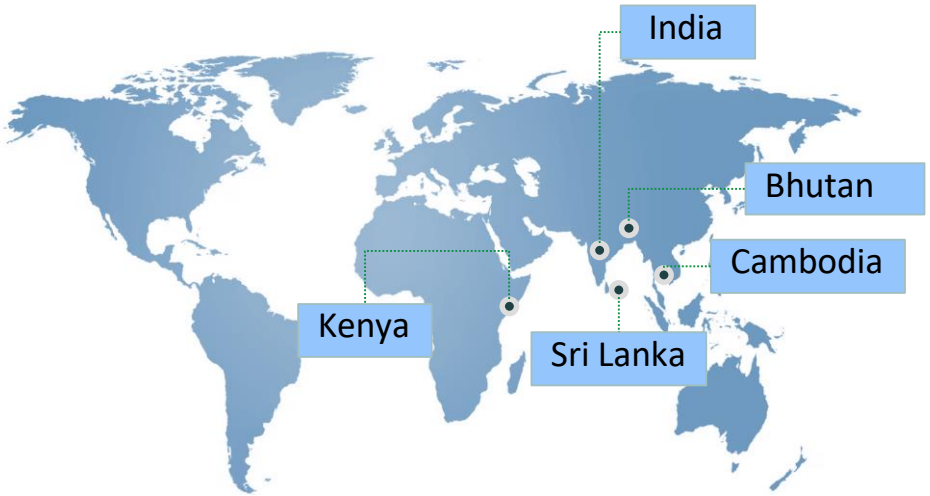
Overview

1. Data Networks, is a provider of innovative software Solutions and Services for the Banking and Financial services industry.

2. With a mission to drive digital transformation, we empower banks, credit unions, and other financial institutions to enhance customer experiences, streamline operations, and maintain regulatory compliance.

3. Our cutting-edge solution is designed to meet the unique needs of the banking sector, enabling seamless integration, robust security, and scalability.

Our Presence



Why We Exist

Vision

To be a trusted partner in the banking sector's digital journey, fostering innovation, security, and agility across financial institutions worldwide.

Mission

To deliver secure, scalable, and user-centric banking software solutions that enable financial institutions to adapt to a rapidly evolving digital landscape and offer superior services to their clients.

Introduction

Bhutan Insurance Limited (BIL) is scaling digitally and requires a robust Early Warning System (EWS) that proactively detects policy-level risks, customer risks, fraudulent behaviour, operational anomalies, claim risk patterns, agent risk indicators, and financial stress signals.

Data Networks proposes a centralized, AI-enabled EWS platform built on strong data integration, rules-based alerts, machine learning scoring models, and real-time dashboards.

Objectives of EWS

The key objectives of the EWS are:

Risk Mitigation

- Identify high-risk customers, policies, agents & claims BEFORE they escalate.
- Reduce claim losses by early fraud detection.

Portfolio Health Monitoring

- Monitor lapse ratio, claim ratio, delayed payments, suspicious patterns.

Improve Operational Efficiency

- Provide actionable alerts to underwriting, claims, compliance & finance teams.

Strengthen Governance

- Maintain compliance with regulatory requirements for insurer-level risk monitoring (similar to RBI/IRDAI governance for banks & insurers).

Scope of Work

Data Networks will develop and deploy an integrated EWS solution covering:

01. Data ingestion from core systems

02. Risk indicator design for insurance

03. Rule-based & ML-based alert generation

04. Scoring engine

05. Case Management System (EWS Action Module)

06. Dashboards & MIS

07. Automated Reporting

08. User access control & audit trail

09. API integrations

10. Deployment, UAT, Training & Go-Live

Functional Requirements

RISK INDICATORS (KRIs – Key Risk Indicators)

Policy Risk

- Address changes
- Policy modifications
- Payment delays

Customer Risk

- Suspicious behavior
- Claim patterns
- KYC issues

Agent Risk

- High claims ratio
- Policy clustering
- Lapse patterns

Claim Risk

- Repeated claims
- Hospital flags
- Amount deviation

Operational Risk

- Approval delays
- Login anomalies
- Override patterns

Risk Indicators

Policy Risk Indicators

- Frequent address changes
- Policy modified multiple times
- Payment bounce / delay
- High-risk geography
- High premium discount
- Short policy duration
- Multiple policies in short time frame

Customer Risk Indicators

- Suspicious customer behaviour
- Claim filed immediately after policy issuance
- Customer with history of large claims
- KYC mismatch
- High-risk occupation

Risk Indicators

Agent / Channel Risk

- Agent with high claim payout ratio
- Agent onboarding abnormal number of high-risk policies
- High lapse ratio
- Suspicious clustering of policies

Claim Risk Indicators

- Repeated claims with same reason
- Claims from network hospitals flagged for fraudulent patterns
- Claim amount deviation vs benchmark
- Suspicious documents or missing data

Risk Indicators

Operational Risk Indicators

- Delayed underwriting approvals
- Multiple overrides by managers
- Suspicious login attempts
- Data modification patterns

EWS Alert Engine

Red Alert

Immediate action required

Orange Alerts

Escalation required

Yellow Alert

Monitoring only

Each alert includes

- Risk Score
- Reason for alert
- Triggered KRIs
- Action required
- TAT for closure
- Assigned team

Scoring Engine

Data Networkswill develop a scoring engine for Risk Rating:

Risk Score Formula

Risk Score =

$$\Sigma\ (\text{KRI Weight} \times \text{KRI Value})$$

Fully configurable by BIL

Risk Categories

Low Risk
Score: 0-30

Low Risk
Score: 0-30

Low Risk
Score: 0-30

Dynamic Risk Scoring

0 — 30 — 60 — 100

0-30	31-60	61-100
Low	Medium	High

Weighted scoring model ensures accurate risk assessment based on multiple KRIs

Case Management System

Every Alert Becomes a Case

Complete traceability and accountability for all risk alerts

Case Features

- Assign to user/team
- Upload evidence
- Add review notes
- Attach documents
- Risk re-evaluation
- Close with status

Case Status Options

- True Risk
- False Positive
- Pending

Full traceability and audit trail for compliance

100%	Real-time	Multi-user
Traceability	Updates	Collaboration

Dashboards & Analytics

Executive Dashboard

- Total Alerts by Severity
- High Risk Policies
- Top Risk Policies
- Geography Heatmap

Claims Dashboard

- Abnormal claim patterns
- Fraud-prone hospitals
- Claim turnaround anomalies

Underwriting Dashboard

- High-risk proposals
- Policy churn
- Lapse probability

Agent Monitoring

- Agent performance
- Risk contribution
- Cancellation clusters

Operational Dashboard

- Underwriting delays
- Login anomalies
- Exception overrides

Reports & Documentation

Daily/Weekly/Monthly EWS Reports

High-risk Customer List

Agent Watch list

Suspicious Claim Report

Audit Trail Report

Export Formats & Delivery

PDF

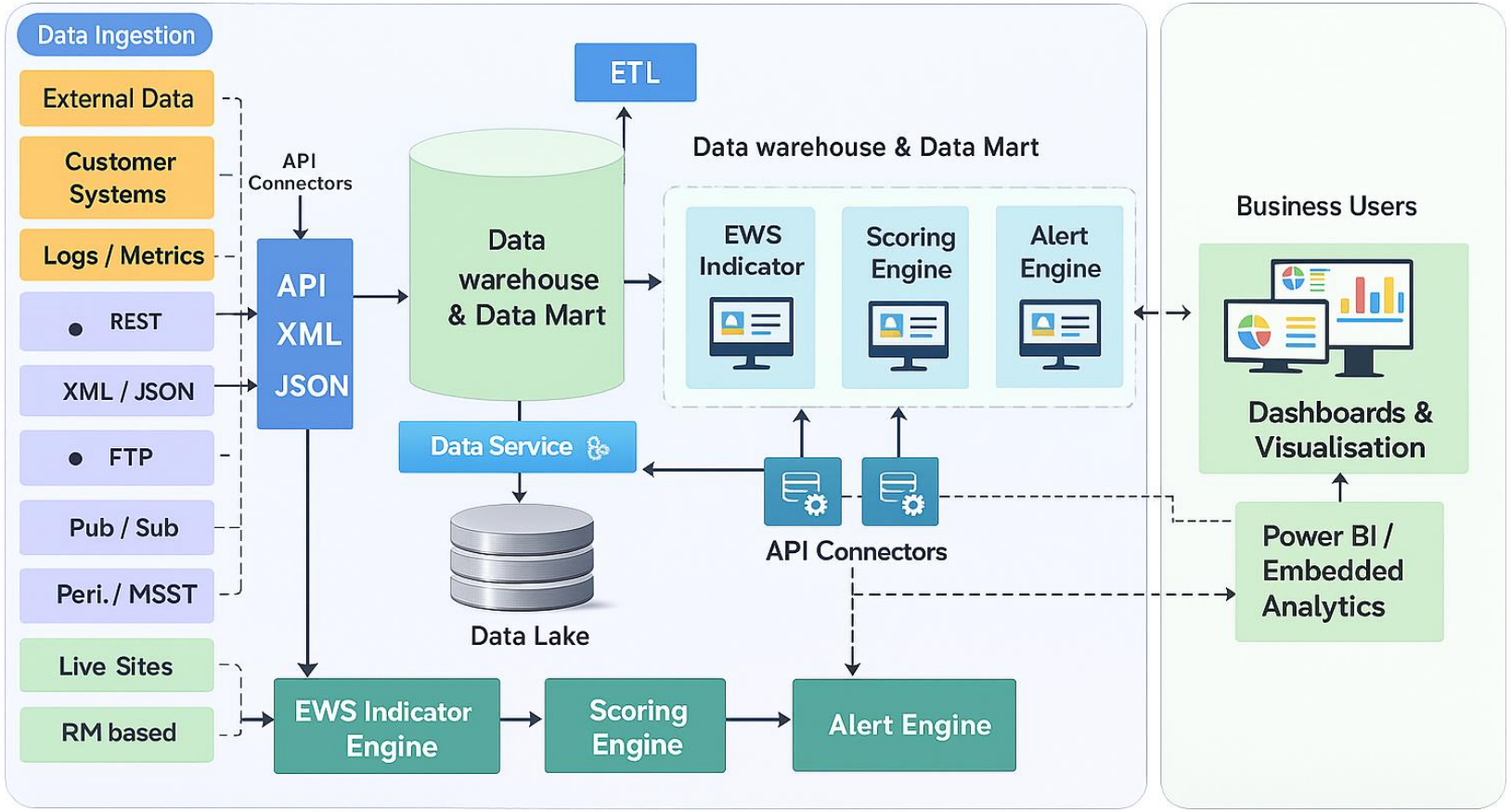
Excel

Email

Layered Architecture

Layers

- ✓ Data Ingestion Layer (ETL / APIs)
- ✓ Data Warehouse / Data Lake
- ✓ EWS Indicator Engine
- ✓ Scoring Engine
- ✓ Alert Engine
- ✓ Dashboards & Visualisation (Power BI / Embedded Analytics)
- ✓ Case Management Layer



Technology Stack

Proposed Technologies

Backend

- .NET Core / Java Spring Boot
- Python for scoring models

Frontend

- Angular / React
- Responsive UI

Database

- PostgreSQL / MS SQL
- Optional: Data Warehouse on AWS Redshift / Azure Synapse

Integration

- REST APIs
- SFTP ingestion
- Cron-based ETL jobs

BI Layer

- Power BI / Metabase / Tableau

Integration Points with BIL Systems

We will integrate with:

- Core Insurance System
- Policy Master
- Claims Management System
- Agent System
- Finance (Payments / Receipts)
- Document Management System
- HR System (for login risk indicators)

Security & Compliance

Role-based Access Control (RBAC)

MFA & Login Audit Trail

Data Encryption (AES-256)

Secure API Gateway

Audit logs for all user actions

BIL-Compliant data retention policy

Why Choose Data Networks?

Industry Expertise

Extensive experience in developing and deploying banking solutions, since 1998, with good knowledge of financial regulations and standards.

Customer-Centric Approach

Focus on user-friendly interfaces and customer experience to help banks foster trust and loyalty.

Scalability and Flexibility

Modular solutions that can grow with your organization, adapting to market changes and technological advancements.

Security Focused

Our solutions are built with security as a priority, with end-to-end encryption, secure access controls, and regular security audits.

Dedicated Support

Comprehensive support teams and a client success program ensure smooth deployment and ongoing maintenance.

DATA NETWORKS

Contact Us

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